

Working master

TOROIDAL — Working Master

Current reference · consolidation v0.3 · Updated: Sep 8, 2026

Author: Anonymous · CC0 1.0 · Mathematical reference and engineering validation

Start here for the toroidal project. This tab consolidates the current definitions, later dynamics result, validation status, and next work. The Earlier geometry release tab preserves the previous document. Detailed proofs, frozen protocols, and execution receipts remain linked sources; their distinct kernels, cohorts, and evidence limits remain operative.

This consolidation replaces the need to navigate several overlapping summaries. It does not replace a source protocol with enough information to execute a run, authorize production, or treat an incomplete code listing as a reproducible package.

1. Current position

The geometry contract is established on its stated domain: integer cut winding for divergence-free currents; modular cut flux for charge-six sources; mod-two membrane constraints; and a separate circular-throat kinematic model.

The TD-COS-FH-001 companion specifies a complete ideal full-state kernel. Its sector observation has no exact finite stationary Markov order at any fixed positive integer spacing of attempted microticks. The proof also excludes an exactly time-homogeneous finite-order sector law from another initial full-state distribution.

The separate L=3 direct-reference follow-up reports engineering-validation PASS. The original L=3 pilot remains UNRESOLVED. Physical Q1/Q2/Q3 are not established here; physical Q2 remains NOT_RUN. Recorded production NOT_AUTHORIZED status is unchanged.

The historical accepted-event discrepancy remains $-1,352$. The all-orders proof is readable, but its embedded verifier ends at “assert a”. Those reproduction gaps remain open.

2. Model, observables, and ensembles

The chosen field constructor has local sign identification $z \sim -z$ and physical composite $\Phi=z^2$. An onsite monomial $z^p\bar{z}^r$ is invariant when $p-r$ is divisible by six. The leading phase-dependent

onsite anisotropy therefore has degree six. This symmetry classification does not derive a unique continuum theory or establish XY* criticality.

Use positively oriented links on a periodic cubical lattice, integer current I , mod-two membrane M , sector $q \in (\mathbb{Z}/2)^3$, and source n . Boundary is head minus tail; physical divergence is $-\partial I$. The consolidated constraints are $\text{div } I + 6n = 0$ and $\bar{I} + \partial M + \sum \alpha q \alpha \Gamma \alpha = 0$ over mod two. Reference cycles $\Gamma \alpha$ and cuts are fixed.

For $F\alpha(k) = \sum x: x\alpha = k \mid x, \alpha$, zero divergence gives $W\alpha = F\alpha(k) = L\alpha^{-1} \sum \ell // \alpha \ell \in \mathbb{Z}$, independent of the parallel cut. Preserve signed W and ΔW . If divergence is nonzero with charge-six sources, cuts agree modulo six; use $a_6 = [I \bmod 6]$ and $q = a_6 \bmod 2$. Store integer winding as unavailable and averaged current separately. Averaged current can be fractional or have the wrong parity.

Let h label fixed-reference Wilson-loop signs, Z_h the direct partition functions, and $\mathfrak{Z}q$ the nonnegative dual coefficients. $Z_h = (1/8) \sum q (-1)^{h \cdot q} \mathfrak{Z}q$ and $\mathfrak{Z}q = \sum h (-1)^{h \cdot q} Z_h$. Consequently $Z_{\text{full}} = \mathfrak{Z}000$, whereas $Z_{000} = (1/8) \sum q \mathfrak{Z}q$. The full periodic ensemble's forced even parity is not confinement evidence.

Canonical Q2 remains the source-free, cubic, fixed-reference Z_{000} branch. Under the stated positivity assumptions, $Z_h/Z_{000} = \text{Ep}[(-1)^{h \cdot q}]$ lies in $[0, 1]$, $p(0) \geq p(q)$, and $p(0) \geq 1/8$. Finite estimates still need uncertainty analysis. Fixed-reference loops do not imply path-independent holonomy.

3. Declared dynamics and exact sector memory

TD-COS-FH-001 uses $X = (I, M, q)$, unbounded integer currents, $J=1$, $t=1/2$, $h_6=0$, and target weight proportional to $t^{\text{NM}} \prod |\mathcal{L}| |\mathcal{I}| (1)$, subject to the constraints. $\mathcal{F}n$ is the modified Bessel function.

At volume V , each attempted microtick chooses among identity, cube, coupled plaquette, even plaquette, closed sheet, even reference cycle, and unit sector cycle, with respective weights $(V, V, 3V, 3V, 1, 3, 3)$ divided by $B=8V+7$. Metropolis acceptance uses the exact target-weight ratio and symmetric descriptor proposals. Identity moves and rejections advance the clock. A sweep is K^B , a fixed number of random-mixture attempts, not one deterministic pass through the families.

On the positive unbounded domain, the specified kernel is reversible, irreducible, aperiodic, and has a normalized positive invariant law. These analytical properties do not establish implementation conformance, a mixing-time bound, or adequacy of the recorded warmup.

For every fixed cubic $L \geq 2$ and fixed integer $d \geq 1$, the stationary observation $Y_n = q(X_{dn})$ is not Markov of any finite order. Finite order would force $A^2 u = c A u$, where $R = K^d$, $u = 1_{q=000}$, D multiplies by u , and $A = DRD$. Valid states with current $2N$ on every positive edge contradict that identity through a nonzero Bessel-tail correction to the return law. The source supplies the analytical argument for all orders; six finite arithmetic cases are supplementary checks.

The theorem applies to the ideal cosine kernel and attempted-microtick spacings, including prescribed sweeps. It does not cover finite-current truncations, accepted-move clocks, adaptive schedules, or every possible augmented statistic. The earlier (q, W) one-step insufficiency is a separate result; no all-orders theorem for (q, W) is asserted.

The exact full state remains Markov. Given an initial law, the causal posterior $v_n(x) = \Pr(X_{dn} = x \mid Y_0, \dots, Y_n)$ is a sufficient belief state, updated by prediction with K^d and conditioning on the next sector. This may require an unbounded representation. Finite practical compression needs its own approximation guarantee.

4. Geometry of discarded information

Fix a domain D , projection $\pi: D \rightarrow Q$, and witness $W: D \rightarrow Y$. Use attained labels $Q = \pi(D)$. The non-descent locus is $NW(\pi) = \{q: \text{there exist } x, y \text{ in } \pi^{-1}(q) \text{ with } W(x) \neq W(y)\}$. The sector space $SW = \text{im}(\pi, W)$ records the witness values actually carried by each fiber.

The refinement $x \mapsto (\pi(x), W(x))$ is the coarsest exact refinement retaining both values. It is a formal witness repair, not automatically an available causal measurement or a minimal mechanistic state. A static recovered witness need not close its own future dynamics.

For prediction, choose $W(x)$ to be the entire next-observation law, not one sampled successor. Equal retained labels must give equal next-label laws for universal one-step closure. With time-dependent updates, test equality within each time slice and across times separately; adding a clock only repairs the latter kind of failure.

For a fixed output and clock, let $\delta h(z)$ be the supremum total-variation distance between h -step output laws from full states sharing retained record z . This diameter is nondecreasing with horizon; refining the retained record cannot increase its corresponding fiber diameter. Any single law used for every state in a fiber has worst-case error at least half that diameter. An exhibited pair supplies a lower bound, not a uniform upper bound.

Before an empirical comparison, freeze the domain, witness, projection, same-fiber rule, metric, tolerance, uncertainty treatment, and coverage. A pair whose distance lower bound exceeds tolerance certifies non-descent at that observed fiber. No detected pair is not a global descent proof.

The bare non-descent definition permits arbitrary shape. Smoothness, connectedness, boundaries, and holonomy require additional hypotheses and observations. For the finite q-space, continuous boundaries require a separately defined parameter or refined base. Declare a path-lifting rule or connection before transport calculations. Chemistry freeze, unobserved global shape, connectivity, and holonomy remain unchanged/unresolved.

4.1 Typed sectors: static loss and predictive memory

This integrates the supplied Toroidal Sectors Defect v0.1 with corrections to its use of Typed Defects v0.2.

On the zero-source unbounded-current domain, write $\pi_q(X)=q$ and $W(X)$ for signed cut winding. Then $\pi_q=\text{parity} \circ W$, so $N_q(W)=\emptyset$. Conversely $N_W(\pi_q)$ is all eight attained sectors: $I=\sum \alpha(q\alpha+2k\alpha)\Gamma\alpha$, $M=0$ satisfies the constraints for every $k \in \mathbb{Z}^3$, and gives $W=q+2k$. Equivalently, add $2\Gamma\alpha$ to any valid representative; divergence and mod-two constraints remain unchanged while signed winding changes by two. This proves static loss of W from q without using the memory theorem.

Membranes are another distinct witness. Adding a nonzero cube boundary to M preserves ∂M and q but changes M . This is a discrete same-sector collision. Calling it differential Kind IV would additionally require the smooth-fiber assumptions from New geometry §16. Restricting a domain is D_4 ; switching to the $L=3$ rotor/gauge kernel is a change of kernel, not an application of D_4 alone. Moving from zero-source to sourced states changes domains and replaces integer winding by modular cut flux.

4.2 The correct history-level reading

For $Y_n=q(X_{dn})$, fix the full-state initial law and $R=K^d$. Let $H_n=(Y_0, \dots, Y_n)$, let $s_m(H_n)$ be its last m symbols, and let $G_n(H_n)=\text{Law}(Y_{n+1} \mid H_n)$ on positive-probability histories. Finite order m requires G_n to factor through s_m at all applicable n , with a common reduced transition rule when time homogeneity is asserted. For stationary infinite pasts, the corresponding condition is equality almost surely of the conditional law given the entire past and that given the last m symbols.

Comparing hidden states with the same finite history tests a stronger, different condition. Failure of strong lumpability alone does not establish infinite Markov order for an initialized observed process. Use the existing TD-COS-FH-001 all-orders proof for the latter conclusion; the supplied increment's hidden-state-history locus is not by itself an equivalent restatement.

4.3 Causal filter: sufficient, not automatically coarsest

With a specified initial law, predict $v_{-(n+1)}(x') = \sum x v_{-n}(x) R(x, x')$. After observing y , set $v_{-(n+1)}(x') = 1_{\{q(x')=y\}} v_{-(n+1)}(x') / \sum z: q(z)=y v_{-(n+1)}(z)$, whenever the denominator is positive. This causal filter determines future sector laws without reading future observations.

The map from a posterior to its future-sector law need not be injective. Therefore v_{-n} is a sufficient belief state, not automatically S_W or the coarsest predictive refinement. The coarsest history refinement retaining a declared current label and predictive law is their attained pair. Finite compression must be tested for its own exact or approximate loss.

A finite compression can create a fiber collision, or only a bounded approximation error; neither is automatically a Kind V representability gap. Kind V requires an identified object outside a specified representation image.

4.4 Discrete topology and status

Every subset of $Q = (\mathbb{Z}/2)^3$ is clopen and has empty topological boundary in its discrete topology. Smooth transport requires extra structure; discrete graph transport or gauge holonomy requires its own definitions and is not excluded merely because the labels are discrete.

These corrections change the integration language, not TD-COS-FH-001, the separate (q, W) one-step result, or source evidence. L=3 engineering PASS, original-pilot UNRESOLVED, physical Q2 NOT_RUN, and the historical -1,352 accepted-event discrepancy remain as recorded. No new replay or physical run is claimed.

Corrected calculus, §16: [New geometry](#)

4.5 Spectral imports require a declared model

The corrected v2.2 increment defines spectral family VI on operator or path domains and proves two exact examples there. A toroidal application would need an explicit state-to-operator or state-to-path assignment, its analytical conventions, and a witness tested against a declared observation. Naming η or a boundary operator alone supplies no non-descent certificate.

No such identification with K , signed winding, or the membrane witness is established here. A continuum limit is not assumed. The discrete membrane collision in §4.1 retains its corrected status, and the TD-COS-FH-001 all-orders proof keeps its separate history-level argument.

Spectral examples and foreign-import table: [GEOMETRY — Upgrade v2.2grok](#)

5. Smooth-phase algebra: corrected interface

The smooth phase model remains a separate mathematical object. Use $B_h = C^\infty(\mathbb{T}; \mathbb{C})$ and $D_h = (2\pi i h')^{-1} \partial_x$ for an orientation-preserving smooth circle coordinate h with $h' > 0$.

The products $f \circ g = fDhg$ and $f \triangleleft g = gDhf$ are opposite Novikov products. Their associators are $-fgDh^2u$ and $guDh^2f$. The bracket is $[f,g]D = fDhg - gDhf$; the opposite product reverses its commutator sign.

Phase inclusions of degrees $k=13$ and 39 satisfy $DhJk = kJkD$. Their intrinsic normalized derivatives are Dh/k . Finite-fiber averaging is not a product homomorphism: $E(e1) = E(e-1) = 0$ but $E(e1 \circ e-1) = -1$.

The identity $(fg) \circ u = f(g \circ u)$ holds even in an associative noncommutative algebra for this recipe. The actual rotation-algebra Novikov defect is $(\ell^{-1} - 1)UV^2$ when $UV = \ell VU$; its trace can vanish while its squared trace norm is positive.

The recovered GD embedding uses $Bh[t, t^{-1}]$, $d = t^{-1}Dh$, $\{A, B\} = AtDhB - DhA Bt$, and $\iota(f) = tf$. It preserves the two GD operations; it does not turn the original pointwise Wronskian bracket into a Poisson bracket.

The §35 audit reports 36,237 exact assertions, including expected nonzero defects, in an independent implementation. This consolidation has not rerun that verifier. These are smooth-phase algebra results, with no inferred physical Hamiltonian, central charge, or Q2 verdict.

6. Validation, signed winding, and decisions

The separate $L=3$ direct-reference follow-up used eight fresh seeds 196000–196007, 2,000 warmup plus 40,000 retained sweeps per chain, for 320,000 retained states. It reports replay matching all retained states and 336 state/RNG checkpoints. This direct rotor/gauge sampler has its own sweep; it is not the attempted-microtick kernel used in the memory theorem.

With encoding $h = hx + 2hy + 4hz$, follow-up counts are [307471, 3858, 4054, 217, 3798, 265, 233, 104]. Every bin meets the original raw-count minimum 100. All eight approximate engineering comparison screens pass against the preserved 32,000 dual samples. The original pilot's minimum 33 remains UNRESOLVED; cohorts are not pooled.

The direct-to-dual reference is $\hat{p}(q) = \sum h(-1)^h \cdot qHh / (8H000)$. The frozen screen is $|dual-reference| + 5 SE_{combined} \leq 0.04$ for every sector, with the source's chain-based uncertainty procedure. Raw count is not effective sample size; these screens are not an exact simultaneous-confidence theorem or proof of production mixing.

In this review, the count sum, minimum, eight Fourier-reference probabilities, and six exact character-return/convolution cases were independently checked. The long simulation runs and complete §35 verifier were not rerun.

SW1 retains signed integer winding only on its divergence-free domain. Under a current-reversal-invariant target, $\text{Sodd}, \alpha = E[q\alpha \text{sgn } W\alpha] = 0$, and $\text{fodd}\pm = (\text{fodd}\pm \text{Sodd})/2$. This supplemental check cannot replace a frozen primary score. Physical SW1 remains NOT_RUN in the attached release.

Keep fit validity, scale admissibility, accessibility, observed coverage, stationarity, and mixing separate. The heuristic $L_{\max} \geq 4\xi_{\text{UCB}}$ applies only to an admitted finite-radius branch. Missing or discordant fits remain unresolved. A long-range preference makes the finite-radius rule inapplicable and requires a separate frozen scientific verdict path.

For a conditional claim $A \Rightarrow B$, failed validity prevents an admissible verdict; unresolved required inputs leave the joint claim unresolved. A failed antecedent rejects that realization. All antecedents satisfied with B failing falsifies that specified conditional prediction for that realization. Software checks cannot supply missing physical antecedents.

7. Throat and two-clock comparators

The circular throat is a continuum kinematic comparator with centerline radius A and metric $h^2 ds^2 + dr^2 + r^2 d\vartheta^2$, $h = 1 - (r/A)\cos\vartheta > 0$. Physical components $v_s = U(s)$, $v_r = -rU'(s)/(2h)$, $v_\vartheta = 0$ are divergence-free. Require positive U , circle-periodic U , and tube radii below A .

The material boundary $R_{\text{mat}} = R_0 \sqrt{(U_0/U)}$ carries constant flow $\pi R_0^2 U_0$ with no relative side flux. A fixed-radius control tube has $Q_c = \pi R_c^2 U$ and outward side flux $Q_c(a) - Q_c(b)$. These are different boundaries. No gauge-force law, physical real-time flow, or empirical throat validation follows.

For the separate rotating-lattice comparator, use m sites and relative phase $x_\square = \text{frac}(m[\theta_\square - g(t_\square)])$. Its nearest-site distance is $\|x_\square\|/m$. At regular spacing Δt without resets, $x_{\square+i} = \text{frac}(x_\square + \beta)$, $\beta = m(\alpha + \Delta t/T_g) \bmod 1$. Retaining distance alone generally loses predictive state.

The two-clock specification remains prospective and NOT_RUN, with timing/reset, eligibility, outcome, holdout, and multiplicity inputs unresolved. Historical rotating-lattice nulls remain unchanged. Exact phase arithmetic, spatial cocycles, stochastic sector dynamics, and observed physical behavior remain separate typed relations.

8. Next work and maintenance

First recover the complete versioned verifier and the sampler/event/checkpoint authority needed for replay. Preserve the 1,107,368 versus 1,108,720 accepted-event counts as a historical discrepancy. If recovery is impossible, use the separately dated replacement amendment to define a new replay target; do not silently alter the old kernel or declare the old discrepancy resolved.

Then evaluate candidate retained coordinates for a declared output, horizon, clock, and error criterion. Use full-state or causal-filter evidence, preserve uncertainty and coverage, and distinguish exact obstruction from useful approximate prediction.

Keep this document as the active toroidal reference. Update its relevant section when work changes. Create separate records only for a new proof, frozen protocol, dataset, or execution receipt with its own identity; place those behind the source links. The older release tab is reference history, not a competing current summary.

9. Source documents

Declared cosine kernel and numerical policy: [# Toroidal Dynamics v0](#)

All-orders sector-memory proof:

[# TD-COS-FH-001- the sector process has no finite Markov order](#)

L=3 direct-reference follow-up and its screens:

[# Tiny-system validation — L=3 direct-reference follow-up v0](#)

Original L=3 pilot and extension: [# Tiny-system validation — L=3 extension and pilot v0](#)

Historical mismatch investigation:

[# TD-COS-FH-001 — deterministic mismatch investigation v0](#)

Prospective replacement amendment: [# Prospective replacement amendment](#)

Recovered §35 definitions and algebra audit: [# Toroidal note versus recovered §35](#)

Non-descent geometry and empirical rules: [# New geometry](#)

Finite-horizon predictive loss and phase baseline: [# Geometry Maximization v2](#)

Hidden Quotient witness and residual corrections:

[# THE HIDDEN QUOTIENT — CORE, WITNESS, AND STRIKE](#)

Field definitions and lineage: [# Field Theory Update v0](#)

Protocol reference and execution boundaries:  # Simulation Protocol v0.5

Full source archive:

https://drive.google.com/drive/folders/17P1St3hK26XIWkgSijHamP5bmeyO_weg

Earlier TOROIDAL MASTER v0.2, with detailed derivations:

https://drive.google.com/file/d/1103tr4UUEIbyLYUxfE1zaCyn_E6MyAxW/view

Earlier geometry release

Current integration: declared dynamics, memory, and validation

Updated: Sep 8, 2026

This update records the later TD-COS-FH-001 companions alongside the v1.0 static geometry. The source-free integer winding, charge-six modular cut flux, membrane constraints, and curved-throat kinematics below keep their stated domains. The dynamics result in this update uses its separate zero-source kernel.

The sampler is now specified in a companion

Toroidal Dynamics v0.1 declares $X=(I,M,q)$, the constrained unbounded-current state space, the target weight proportional to $\prod_l I_l |I_l| (1) \cdot t^{|M|}$ at $t=1/2$, and the proposal/acceptance schedule with $B=8L^3+7$. Here $I_n(1)$ denotes the modified Bessel function. Identity proposals and rejected proposals both advance attempted microticks. This supplies a mathematical dynamics specification; it does not by itself reproduce an executable run.

Static sector validity does not give predictive closure

For that TD-COS-FH-001 kernel at $J=1$, $t=1/2$, $h_6=0$, fixed reference Z_{000} , and cubic $L \geq 2$, the observed process $q(X_{dn})$ has no exact finite Markov order at any fixed positive integer attempted-microtick spacing d . The written proof applies to unbounded integer currents. It shows that a finite sector history cannot replace the full predictive state exactly, even though each sector value is a valid topological observation.

The theorem is a law-level statement, separate from the finite geometric fixtures in §2. It does not certify a finite-current implementation, accepted-move clock, adaptive schedule, physical confinement, or production mixing. Finite-dimensional hidden-state representations are not all ruled out. A posterior over the declared full state supplies a formal belief-state route; a practical compression needs a measured or bounded approximation error.

$L=3$ follow-up: reported PASS, with the original pilot kept separate

The direct-reference follow-up reports eight fresh direct chains, seeds 196000–196007, each with 2,000 warmup sweeps and 40,000 retained sweeps: 320,000 retained states total. Its reported minimum holonomy count is 104, meeting the threshold 100, and all eight Fourier-sector comparison screens pass against the preserved dual reference. The cohort was fixed before that follow-up ran, but its local time seal is not an external preregistration anchor.

With direct-holonomy encoding $hx+2hy+4hz$, the counts are [307471, 3858, 4054, 217, 3798, 265, 233, 104], which sum to 320,000. The original pilot's undercovered result remains UNRESOLVED; its minimum count 33 is not retroactively changed and the cohorts are not pooled. Physical Q2 remains NOT_RUN. Count coverage does not estimate effective sample size or establish production mixing.

Reproduction work still open

The deterministic investigation retains an accepted-event mismatch: 1,107,368 versus 1,108,720, a difference of $-1,352$. A matching short prefix does not resolve the full-run discrepancy. The all-orders Doc also ends mid-verifier at “assert a”; its complete executable needs recovery. Do not alter the mathematical kernel simply to force agreement with an unexplained count.

The present review independently checked the count/Fourier arithmetic and six exact finite-group character-return cases. It did not replay the reported 320,000-state runs or validate production behavior. The next reproducibility step is to recover the versioned executable and event trace, reconcile the mismatch under the fixed attempted-microtick schedule, and report any changed scientific status in its own frozen protocol.

Linked sources

Toroidal Dynamics v0.1 specification: [# Toroidal Dynamics v0](#)

No finite Markov order proof:

[# TD-COS-FH-001- the sector process has no finite Markov order](#)

L=3 direct-reference follow-up: [# Tiny-system validation — L=3 direct-reference follow-up v0](#)

Accepted-event mismatch investigation:

[# TD-COS-FH-001 — deterministic mismatch investigation v0](#)

Document body

Toroidal Geometry v1.0

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A coherent successor to the toroidal packet, grounded in Geometry Maximization v2.0

6 September 2026 · Mathematical reference release

The geometry now has an executable consistency check. A supplied integer current, mod-two membrane and sector label can be checked on a rectangular three-torus. The throat construction now includes the metric and velocity correction required by actual curvature. The field, protocol, appendices and evidence records use these same definitions.

1. The decisive correction

The signed flux through a cut is

$$\begin{aligned} & \backslash \\ F_{\alpha}(k) &= \sum_{x: x_{\alpha}=k} I_{\{x, \alpha\}}. \\ & \backslash \end{aligned}$$

If the integer current is divergence-free, every parallel cut gives the same integer winding. On a rectangular torus,

$$\begin{aligned} & \backslash \\ W_{\alpha} &= F_{\alpha}(k) = \frac{1}{L_{\alpha}} \sum_{\ell \parallel \alpha} I_{\ell}. \\ & \backslash \end{aligned}$$

With charge-six sources, cuts instead agree modulo six. The correct sector is the parity of that modular cut flux. The volume average can be fractional, or an integer with the wrong parity. The included examples exhibit both failures and accept the underlying sourced states as geometrically valid.

The validator therefore reports ``integer_winding: null`` when integer divergence is nonzero. It still reports the actual cuts, modular flux, parity and volume average under separate names. At zero divergence, signed winding is retained even when it is negative or changes by an even integer.

2. What is runnable

| Component | Implemented result | Scope |

|---|---|---|

| Integer cubical chains | Boundary, divergence, cut flux and state constraints | Exact finite arithmetic |

| Rectangular topology | Boundary ranks and Betti numbers over mod two | Three tested shapes; reusable bounded validator |

| Operator records | Cube, plaquette, sheet, even-cycle, signed sector and rejected updates | Constructed outcomes with a complete eleven-event ledger |

| Modular currents | Cut classes for moduli 2, 3, 4, 6 and 8 | General geometry; the gauge-state contract requires even modulus |

| Accessibility and closure | Exact finite graph/kernel counterexamples | Named finite fixtures |

| Character transform | Exact positive coefficient example and inverse | Algebraic fixture; no microscopic partition sum enumeration |

| Curved throat | Metric, corrected velocity, flux and tangency checks | Circular continuum kinematics |
 | Reporting rules | Scale and conditional-verdict tables | Decision logic, without fitted data |

Measured outcomes are in [the verification receipt](VERIFICATION_RESULTS.json). The suite checks the boundary-square identity over every cell of three shapes, rejects inconsistent states, checks all 64 diagonal-routing rectangles with sides 2 through 9, and distinguishes weak stationary closure from universal closure. Numerical throat checks include an independent Cartesian divergence calculation and integration over the curved side surface.

3. The curved-throat result

Let the centerline be a circle of radius $\backslash(A\backslash)$, with arclength $\backslash(s\backslash)$, cross-sectional radius $\backslash(r\backslash)$, angle $\backslash(\vartheta\backslash)$, and

$$\backslash[h=1-\frac{rA\cos\vartheta}{A}>0.$$

$\backslash]$
 The metric is $\backslash(h^2ds^2+dr^2+r^2d\vartheta^2\backslash)$. Physical velocity components

$$\backslash[v_s=U(s),\quad v_r=-\frac{rU'(s)}{2h},\quad v_\vartheta=0$$

$\backslash]$
 are exactly divergence-free. The straight-tube radial velocity lacks the factor $\backslash(1/h\backslash)$ and is generally not divergence-free in this geometry.

The material tube $\backslash(R_{\rm mat}=R_0\sqrt{U_0/U}\backslash)$ has constant through-flow and no relative side flux. A fixed-radius control tube has $\backslash(Q_c=\pi R_c^2U\backslash)$ and outward side flux $\backslash(Q_c(a)-Q_c(b))$ over $\backslash[a,b\backslash)$. Its flux increments add along the circle and sum to zero over a full turn. Those facts coexist because the two boundaries are different. [Full derivation and limits](05_TTSC_1_v0.4_Curved_Throat.md).

![Curved-throat material and fixed-radius profiles](figures/curved_throat.png)

4. The geometry extensions that are justified

Rectangular tori need axis-specific lengths and otherwise retain the same chain and cut arguments. For sources divisible by $\backslash(m\backslash)$, currents define mod- $\backslash(m\backslash)$ cut classes; reduction to parity is a general group map only when $\backslash(m\backslash)$ is even. With multiple allowed source charges, their positive greatest common divisor gives the universally conserved modular class. Divisibility by one gives no nontrivial modular label.

On a rectangular diagonal-routing chart of periods $\backslash(M,N\backslash)$, the reachable subgroup has index $\backslash(\gcd(2,M,N)\backslash)$. There are two sheets precisely when both periods are even. The expression $\backslash((x+y)\bmod 2\backslash)$ is not a globally defined torus label when either period is odd.

The curved model treats a circular centerline with a regular tubular neighborhood. Arbitrary centerline curvature, torsion, non-orientable spaces and dynamical force balance require additional derivations and are outside this release.

5. Evidence carried forward

The phase-screen mathematics and explorer remain in the intact v2.0 baseline. The 22-call and separate 40-call rotating-lattice records retain their reported failed primary gates. The later arriving-lag lead remains prospective. Unified Return closure remains unresolved; the Pell empirical bridge remains unestablished. These are preserved source assessments, not newly recomputed experiments.

No toroidal production sampler, acceptance law, equilibrium evidence, Q1/Q2/Q3 result, external provenance anchor, or empirical throat validation is supplied here. The constructed ledger makes the state conventions reviewable; it cannot establish stationarity or mixing. The concrete next empirical task is to specify and validate a full sampler and its schedule, then run the frozen field protocol.

Use [the protocol](02_Simulation_Protocol_v0.6_Consolidated.md) for execution details and [the manifest](15_Integration_Manifest_v0.3.md) for the complete migration map.